Clustering and Comparing Safety in Cycling Networks

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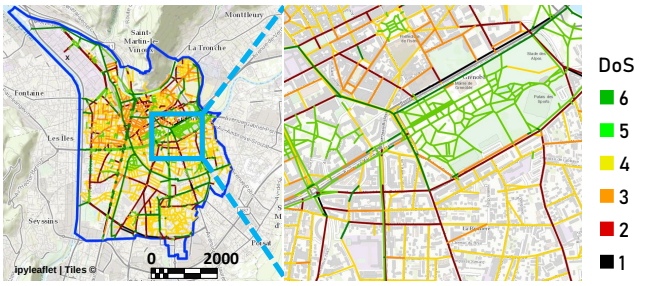


Fig. 1. Cycling network of Grenoble, France. The degree of separation (DoS) score of the cycling infrastructure from motorized vehicles is indicated in ascending order from 1 to 6.

***Abstract*—This paper proposes a safety-aware framework for comparing cycling networks of different sizes by clustering the larger network into a super-network with the same number of nodes as the smaller one. We introduce a constrained spectral k-means algorithm that groups arcs into clusters while preserving key safety attributes: degree of separation from motor traffic, cyclist flow, distance, slope, and angular change. The method controls deviations in total cyclist exposure and flow and always returns a feasible clustering. On the clustered network, we define node-exposure weights and present two complementary distances, a Wasserstein metric between exposure distributions, and an exposure-weighted kernel distance between adjacency matrices. A case study on three French cities (Grenoble, Lyon, Strasbourg) shows that the proposed clustering preserves exposure and flow with limited loss and maintains realistic topological structure. Wasserstein and kernel distances jointly indicate that Strasbourg is more similar to Grenoble than Lyon is, in line with independent perception-based safety rankings. The results suggest that safety-preserving clustering combined with optimal-transport and kernel-based distances provides a practical tool for benchmarking cycling networks and, in future work, for performing scalable safety optimization on clustered networks.**

***Index Terms*—Cycling networks, constrained clustering, cyclist exposure, optimal transport, graph kernel, urban mobility**

I. Introduction

The health, environmental, and socio-economic benefits of cycling are well established, including improved cardiovascular health, reduced emissions and air pollution, lower transportation costs, and decreased traffic congestion [1], [2]. Consequently, public authorities increasingly promote cycling as an alternative to motorized transport for short and medium trips. However, concerns about safety and comfort remain major barriers to cycling adoption [3]. Dedicated infrastructure that separates cyclists from motorized traffic significantly improves perceived safety and comfort and is associated with higher cycling uptake [4], [5]. Well-connected and high-quality cycling networks can substantially increase ridership [6], [7]. Despite this, analyzing real-world cycling networks is challenging due to their size, spatial complexity, and the multiple interacting factors influencing route choice and perceived safety. Moreover, comparing the safety performance of cycling networks across cities in a fair and scale-independent manner remains an open problem, yet is crucial for transferring successful planning strategies between urban contexts.

We propose a mathematical framework to compare cycling-network safety across cities of different sizes and levels of detail. Large road graphs are compressed into cluster-based super-networks using a constrained, safety-aware k-means algorithm on a spectral embedding. This process preserves key attributes such as distance, slope, degree of separation, traffic exposure, flow, and angular change while limiting information loss. On the resulting clustered networks, we construct exposure-based kernels that enable structural similarity comparisons between cities. Experiments on real-world networks show that these representations retain safety-critical structure sufficiently to support consistent and interpretable cross-city comparisons.

Related work: Our contribution builds on two research areas: transport-network clustering and network comparison.

Community detection: Clustering and community detection are widely used to uncover structure in complex networks. Classical k-means [8] minimizes within-cluster variance, while graph-based methods identify densely connected node groups, including edge-betweenness [9], modularity maximization [10], spectral methods [11], and physics-inspired approaches [12]. In transportation research, clustering has been applied to traffic-flow analysis [13], [14], driving-style characterization [15], and temporal pattern discovery [16]. For spatial networks, community detection supports control-region design and equity analysis [17]. In cycling studies, clustering has been used to design bike networks from demand patterns [18] and to improve connectivity under budget constraints [19]. However, existing methods are not explicitly designed to preserve cyclist-oriented safety attributes such as exposure, slope, traffic conditions, or angular change when compressing large networks for cross-city comparison.

Network comparison: Comparing transport networks can be viewed as a relaxed form of graph isomorphism, where structural similarity is quantified by a distance metric [20]. Direct comparisons of adjacency matrices are simple but sensitive to node labeling and local perturbations [21]. More robust approaches rely on structural descriptors, including global statistics, graphlets, role-based features [22], [23], and spectral signatures [24]. Graph kernels embed networks into

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DoS	Description	AADT (veh/day)
6	Dedicated bike lanes, fully separated from motor traffic.	14 400
5	Bike lanes shared with pedestrians, separate from cars.	16 000
4	Mixed-traffic roads (30 km/h) without designated bike lanes.	960
3	Shared roads with cycling pictograms.	3 120
2	Roads with painted bike lanes.	6 912
1	Unmarked shared car-bike lanes.	8 448

TABLE I
THE DEGREE OF SEPARATION (DOS) CATEGORIZES BIKE LANE INFRASTRUCTURE BASED ON SAFETY LEVELS. THE THIRD COLUMN PRESENTS ANNUAL AVERAGE DAILY TRAFFIC (AADT) ON ADJACENT ROADS, SOURCED FROM THE GTLVILLE PLATFORM IN GRENOBLE [31].

reproducing kernel Hilbert spaces, enabling similarity comparisons via random walks or marginalized feature spaces [25]–[27]. Optimal transport theory provides a complementary perspective, with Wasserstein distances offering meaningful metrics between distributions [28]–[30]. In this work, this framework underpins the comparison of exposure distributions on clustered cycling networks. However, existing methods rarely account for safety-specific attributes or jointly address multi-attribute exposure and differences in network size when comparing cities.

Statement of Contribution: We introduce a mathematical framework that clusters large cycling networks while explicitly preserving a composite safety indicator based on cyclist flow, traffic exposure, slope, and angular change. The resulting supernetwork representations minimize distortion of safety attributes and reduce all cities to graphs with identical node cardinality. This enables fair, scale-independent comparison of cycling-network safety across cities using exposure-based distance measures.

II. MODEL DESIGN

In this section, we first present the bike lane classification and the graph representation. Next, we introduce the cyclist flow prediction model and the bi-level optimization problems: The first adopts a utilitarian perspective to minimize cyclists' traffic exposure, directing flows toward safer routes, and reducing high-traffic exposure to improve overall safety, while the second follows a sufficientarian perspective, aiming to reduce the number of cyclists falling below relative safety thresholds. Finally, we introduce the two algorithms designed to solve these NP-hard problems.

A. Bike Lane Safety classification

The DoS classification, adapted from an OSM-based platform, categorizes cycling infrastructure based on separation from motor traffic. Table I presents DoS categories, showing that higher-DoS lanes offer greater safety. These scores serve as weights in the network graph, modeling upgrades from lower to higher safety levels. The classification methodology is detailed in [32]. Fig. 1 visualizes these classifications in the Grenoble cycling network.

B. Graph Representation

Let the network be represented by a weighted directed graph (digraph) $\mathcal{G} = (\mathcal{V}, \mathcal{A}, \mathcal{W})$, where $\mathcal{V}$ is the set of nodes corresponding to intersections or road endpoints, $\mathcal{A}$ is the set of directed arcs representing roads, and $\mathcal{W}$ is the set of weights assigned to those arcs. For each arc $a = (u, v) \in \mathcal{A}$, with $u, v \in \mathcal{V}$, the weight $w_a \in \mathcal{W}$ represents the DoS associated with the link. The weights are assumed to be symmetric, meaning $w_{(u,v)} = w_{(v,u)}$. The geographical length of each arc a is denoted by d_a.

C. Cyclist Flows

Following [33], we use an all-or-nothing (AON) traffic assignment model that assumes no congestion and assigns all demand $q_{(u,v)}$ from the EMD2020 survey [34] to the shortest available path between nodes $u, v \in \mathcal{V}$. This reflects typical cycling dynamics, where congestion effects are minimal. In this section, we formalize and adopt the cyclist distance proposed in [35], which defines this distance using OpenStreetMap and Cardiff cycling flow data while incorporating infrastructure characteristics. Then, betweenness centrality is applied to estimate the flow distribution.

To account for cycling-specific factors, we use a modified distance metric that accounts for key factors influencing route choice. Let $\mathcal{V}$ denote the set of nodes in the network, and let $u, v \in \mathcal{V}$ be two nodes. The shortest path between u and v is represented as $p_{(u,v)} = (a_1, \ldots, a_k)$, where each arc $a \in p_{(u,v)}$ has a geographical length d_a, an angular change $\measuredangle_a$ (in degrees, either along a or at junctions between consecutive arcs), a slope factor s_a (which depends on the terrain slope m_a), and a traffic factor f_a (that takes into account the flow of car traffic on roads adjacent to bike lanes). We define the cyclist distance $b_{(u,v)}$ between nodes u and v as

$$b_{(u,v)} = \sum_{a \in p_{(u,v)}} (d_a \cdot s_a \cdot (f_a)^{\rho}) + \sum_{a \in p_{(u,v)}} \measuredangle_a \cdot \kappa, \quad (1)$$

where $\rho > 0$ and $\kappa > 0$ are calibration parameters controlling the influence of traffic conditions and angular changes, respectively. The traffic factor f_a is defined as

$$f_a = \begin{cases} \psi \cdot \exp\left(x_a^{\text{AADT}}\right) & \text{if } w_a < 5, \\ \psi \cdot 2.03 & \text{if } w_a = 5, \\ \psi & \text{if } w_a = 6. \end{cases} \quad (2)$$

Here, x_a^{AADT} represents the annual average daily traffic (AADT) of the adjacent car lanes, expressed in thousands. The variable w_a denotes the DoS of the bike lane a. According to [36], the impact of annual average daily traffic (AADT) is neutralized when $w_a \geq 5$, indicating full separation from car lanes. The factor 2.03 accounts for an observed increase in crash risk in shared spaces with pedestrians compared to separate, exclusive bicycle lanes [37]. The parameter ψ is a scaling factor. Let $p_{(u,v)}$ be the shortest path between nodes u and v based on the cyclist distance in (1). Since the origin-destination (OD) demand in our data is symmetric, each node

acts as both an origin and a destination, ensuring balanced inflow and outflow. The flow conservation for paths is given by

$$\phi_{p_{(u,v)}} = q_{(u,v)}, \tag{3}$$

where $q_{(u,v)}$ represents the demand between nodes u and v. Next, let $\mathbb{I}_{a\in p_{(u,v)}}$ be a binary variable indicating whether arc a is part of the shortest path $p_{(u,v)}$. Then, the total cyclist flow on arc a, denoted as ϕ_a, is

$$\begin{aligned} \phi_a &= \sum_{u,v\in\mathcal{V}} q_{(u,v)} \cdot \mathbb{I}_{a\in p_{(u,v)}} \geq 0, \quad \forall a \in \mathcal{A} \\ \Phi &= \sum_{a\in\mathcal{A}} \phi_a. \end{aligned} \tag{4}$$

This ensures that the total flow on each arc is the sum of all demands for trips passing through it, maintaining balance across the network due to the symmetry of the demand matrix. To assess cyclist exposure to traffic conditions, we introduce $\mathcal{E}$ as a general measure of exposure, defined for pahts in the network. This metric helps evaluate network improvements. For a shortest path $p_{(u,v)} = (a_1, \ldots, a_k)$, cyclist exposure captures the combined effect of traffic-related risk and the distance of cyclist interaction with traffic, and is defined as

$$\mathcal{E}_{p_{(u,v)}} = \sum_{a\in p_{(u,v)}} d_a \cdot f_a \cdot \phi_a, \tag{5}$$

where ϕ_a, d_a and f_a are the normalized arc flow, length and traffic factor, respectively. For a single arc $a \in \mathcal{A}$, cyclist exposure simplifies to

$$\mathcal{E}_a = d_a \cdot f_a \cdot \phi_a \in [0, 1]. \tag{6}$$

Then, for the original network the total cyclist exposure is:

$$\mathcal{E}^{\text{orig}} = \frac{1}{|\mathcal{A}|} \sum_{(v_i,v_j)\in\mathcal{A}} \phi_{(v_i,v_j)} \cdot f(w_{(v_i,v_j)}) \cdot d_{(v_i,v_j)}. \tag{7}$$

D. Cluster Network Definition

Let $\mathcal{P} = \{C_1, \ldots, C_k\}$ be a partition of the node set $\mathcal{V}$ into k non-empty clusters. We define the clustered network as $\tilde{\mathcal{G}} = (\tilde{\mathcal{V}}, \tilde{\mathcal{A}}, \tilde{\mathcal{W}})$, where the supernode set is $\tilde{\mathcal{V}} = \mathcal{P} = \{C_1, \ldots, C_k\}$. For any pair of clusters $(C_r, C_{r'})$, we define the set of arcs between them as

$$\mathcal{A}(C_r, C_{r'}) \;=\; \big\{(u,v) \in \mathcal{A} \;\big|\; u \in C_r,\; v \in C_{r'}\big\}.$$

We include a superarc $\tilde{a} = (C_r, C_{r'})$ in $\tilde{\mathcal{A}}$ if and only if $\mathcal{A}(C_r, C_{r'}) \neq \varnothing$. The clustered arc weights are split into *inter*- and *intra*-cluster components. For $r \neq r'$, the inter-cluster DoS weight between C_r and $C_{r'}$ is defined as

$$\tilde{w}^{\text{inter}}_{(C_r,C_{r'})} \;=\; \min_{a\in\mathcal{A}(C_r,C_{r'})} w_a, \tag{8}$$

i.e., the minimum DoS (highest-risk) connection between the two clusters. For intra-cluster connections, we write

$$\mathcal{A}(C_r) \;:=\; \mathcal{A}(C_r, C_r) \;=\; \big\{(u,v) \in \mathcal{A} \;\big|\; u \in C_r,\; v \in C_r\big\},$$

and define the intra-cluster DoS weight as

$$\tilde{w}^{\text{intra}}_{C_r} \;=\; \min_{a\in\mathcal{A}(C_r)} w_a, \tag{9}$$

which represents the worst-case (highest-risk) connection within cluster C_r.

To enable flow and exposure analysis on the clustered network $\tilde{\mathcal{G}}$, we need to redefine the all-or-nothing traffic assignment approach for this aggregated representation. This requires establishing new rules for path selection and flow distribution across the clustered structure.

E. Clustered cyclist distance

Let $\tilde{p}_{(C_r,C_{r'})} = (\tilde{a}_1, \ldots, \tilde{a}_m)$ denote a shortest path between clusters C_r and $C_{r'}$ in the clustered network $\tilde{\mathcal{G}}$, where each superarc $\tilde{a}_\ell \in \tilde{p}_{(C_r,C_{r'})}$ connects two clusters. We write $\mathcal{C}(C_r, C_{r'}) \subseteq \tilde{\mathcal{V}}$ for the set of clusters visited along this path (including C_r and $C_{r'}$). The clustered cyclist distance between C_r and $C_{r'}$ is defined as

$$\tilde{b}_{(C_r,C_{r'})} \;=\; \underbrace{\sum_{\tilde{a}\in\tilde{p}_{(C_r,C_{r'})}} \tilde{b}^{\text{inter}}_{\tilde{a}}}_{\text{inter-cluster}} + \underbrace{\sum_{C_q\in\mathcal{C}(C_r,C_{r'})} \tilde{b}^{\text{intra}}_{C_q}}_{\text{intra-cluster}}. \tag{10}$$

For each superarc $\tilde{a} = (C_r, C_{r'}) \in \tilde{\mathcal{A}}$ with $r \neq r'$, the inter-cluster contribution is

$$\tilde{b}^{\text{inter}}_{\tilde{a}} \;=\; \big(\tilde{d}^{\text{inter}}_{\tilde{a}} \cdot \tilde{s}^{\text{inter}}_{\tilde{a}} \cdot (\tilde{f}^{\text{inter}}_{\tilde{a}})^{\rho}\big) \;+\; \tilde{\measuredangle}^{\text{inter}}_{\tilde{a}} \cdot \kappa, \tag{11}$$

where the aggregated distance, slope, traffic factor and angular change are defined from the underlying arcs between C_r and $C_{r'}$:

$$\tilde{d}^{\text{inter}}_{\tilde{a}} = \frac{\sum_{a\in\mathcal{A}(C_r,C_{r'})} d_a \cdot \phi_a}{\sum_{a\in\mathcal{A}(C_r,C_{r'})} \phi_a}, \quad \text{(distance)} \tag{12}$$

$$\tilde{s}^{\text{inter}}_{\tilde{a}} = \frac{\sum_{a\in\mathcal{A}(C_r,C_{r'})} s_a \cdot \phi_a}{\sum_{a\in\mathcal{A}(C_r,C_{r'})} \phi_a}, \quad \text{(slope)} \tag{13}$$

$$\tilde{f}^{\text{inter}}_{\tilde{a}} = f_a\big(\tilde{w}^{\text{inter}}_{(C_r,C_{r'})}\big), \quad \text{(traffic factor)} \tag{14}$$

$$\tilde{\measuredangle}^{\text{inter}}_{\tilde{a}} = \frac{\sum_{a\in\mathcal{A}(C_r,C_{r'})} \measuredangle_a \cdot \phi_a}{\sum_{a\in\mathcal{A}(C_r,C_{r'})} \phi_a}, \quad \text{(angular change)}. \tag{15}$$

For each cluster C_r, we define the set of intra-cluster arcs

$$\mathcal{A}(C_r) \;:=\; \mathcal{A}(C_r, C_r) \;=\; \big\{(u,v) \in \mathcal{A} \;\big|\; u \in C_r,\; v \in C_r\big\}.$$

The intra-cluster contribution of C_r is

$$\tilde{b}^{\text{intra}}_{C_r} \;=\; \big(\tilde{d}^{\text{intra}}_{C_r} \cdot \tilde{s}^{\text{intra}}_{C_r} \cdot (\tilde{f}^{\text{intra}}_{C_r})^{\rho}\big) \;+\; \tilde{\measuredangle}^{\text{intra}}_{C_r} \cdot \kappa, \tag{16}$$

with

$$\tilde{d}^{\text{intra}}_{C_r} = \frac{\sum_{a\in\mathcal{A}(C_r)} d_a \cdot \phi_a}{\sum_{a\in\mathcal{A}(C_r)} \phi_a}, \quad \text{(distance)} \tag{17}$$

$$\tilde{s}^{\text{intra}}_{C_r} = \frac{\sum_{a\in\mathcal{A}(C_r)} s_a \cdot \phi_a}{\sum_{a\in\mathcal{A}(C_r)} \phi_a}, \quad \text{(slope)} \tag{18}$$

$$\tilde{f}^{\text{intra}}_{C_r} = f_a\big(\tilde{w}^{\text{intra}}_{C_r}\big), \quad \text{(traffic factor)} \tag{19}$$

$$\tilde{\measuredangle}^{\text{intra}}_{C_r} = \frac{\sum_{a\in\mathcal{A}(C_r)} \measuredangle_a \cdot \phi_a}{\sum_{a\in\mathcal{A}(C_r)} \phi_a}, \quad \text{(angular change)}. \tag{20}$$

Remark 1. *Our clustering methodology and cyclist-distance formulation are designed to capture essential aspects of cycling behaviour while remaining computationally tractable. Key design choices include:*

- *Using minimum DoS values for inter- and intra-cluster connections to follow a conservative, bottleneck-preserving perspective: a single high-risk segment can dominate perceived safety along a route.*
- *Aggregating distance, slope, traffic factor, and angular change via flow-weighted averages, so that frequently used segments dominate cluster-level characteristics and better reflect typical cyclist experience.*

F. Clustered demand, flow assignment, and exposure

Given the original graph $\mathcal{G}$ and its clustered representation $\tilde{\mathcal{G}}$. The clustered demand between two clusters $C_r, C_{r'} \in \mathcal{C}$ is obtained by aggregating the original OD demands $q_{(u,v)}$ with origins in C_r and destinations in $C_{r'}$:

$$\tilde{q}_{(C_r,C_{r'})} := \sum_{\substack{u\in C_r \\ v\in C_{r'}}} q_{(u,v)} \cdot \mathbb{I}_{(u,v)\in\mathcal{A}}, \tag{21}$$

where $\mathbb{I}_{(u,v)\in\mathcal{A}}$ is an indicator function that ensures only valid arcs in the original graph are considered. For the all-or-nothing assignment on the clustered graph, let $\tilde{p}_{(C_r,C_{r'})}$ denote a shortest path between C_r and $C_{r'}$ in $\tilde{\mathcal{G}}$, computed with respect the clustered cyclist distance. We define the inter-cluster flow on a superarc $\tilde{a} \in \tilde{\mathcal{A}}$ as

$$\tilde{\phi}^{\text{inter}}_{\tilde{a}} := \sum_{(r,r')} \tilde{q}_{(C_r,C_{r'})} \cdot \mathbb{I}_{\tilde{a}\in\tilde{p}_{(C_r,C_{r'})}}, \tag{22}$$

where $\mathbb{I}_{\tilde{a}\in\tilde{p}_{(C_r,C_{r'})}}$ indicates whether the superarc $\tilde{a}$ lies on the shortest path between C_r and $C_{r'}$. Similarly, we define the intra-cluster flow associated with a cluster $C_r \in \mathcal{C}$ as the total clustered demand whose shortest path traverses C_r:

$$\tilde{\phi}^{\text{intra}}_{C_r} := \sum_{(r_1,r_2)} \tilde{q}_{(C_{r_1},C_{r_2})} \cdot \mathbb{I}_{C_r\in\tilde{p}_{(C_{r_1},C_{r_2})}}, \tag{23}$$

where $\mathbb{I}_{C_r\in\tilde{p}_{(C_{r_1},C_{r_2})}}$ is 1 if the path between C_{r_1} and C_{r_2} passes through cluster C_r, and 0 otherwise. With all these, finally, on the clustered network $\tilde{\mathcal{G}}$, we define the total cyclist exposure as the average of the inter- and intra-cluster contributions:

$$\begin{aligned} \mathcal{E}^{\text{clust}} &:= \frac{1}{|\tilde{\mathcal{A}}| + |\mathcal{C}|}\Big(\mathcal{E}^{\text{inter}} + \mathcal{E}^{\text{intra}}\Big), \\ \Phi^{\text{clust}} &:= \Phi^{\text{inter}} + \Phi^{\text{intra}}. \end{aligned} \tag{24}$$

Here, the inter- and intra-cluster exposure terms are

$$\begin{aligned} \mathcal{E}^{\text{inter}} &:= \sum_{\tilde{a}\in\tilde{\mathcal{A}}} \tilde{\phi}^{\text{inter}}_{\tilde{a}} \cdot f^{\text{inter}}_{\tilde{a}} \cdot d^{\text{inter}}_{\tilde{a}}, \\ \mathcal{E}^{\text{intra}} &:= \sum_{r=1}^{k} \tilde{\phi}^{\text{intra}}_{C_r} \cdot f^{\text{intra}}_{C_r} \cdot d^{\text{intra}}_{C_r}, \end{aligned} \tag{25}$$

and the corresponding total flows are

$$\begin{aligned} \Phi^{\text{inter}} &:= \sum_{\tilde{a}\in\tilde{\mathcal{A}}} \tilde{\phi}^{\text{inter}}_{\tilde{a}}, \\ \Phi^{\text{intra}} &:= \sum_{r=1}^{k} \tilde{\phi}^{\text{intra}}_{C_r}. \end{aligned} \tag{26}$$

The quantities $d^{\text{inter}}_{\tilde{a}}, f^{\text{inter}}_{\tilde{a}}$ and $d^{\text{intra}}_{C_r}, f^{\text{intra}}_{C_r}$ are the clustered distance and traffic-factor terms defined in the previous subsection for inter- and intra-cluster connections, respectively.

III. Clustering Algorithm

To enable fair comparison between two cycling networks of different sizes, we cluster the larger network so that its number of supernodes matches the number of nodes in the smaller one. The construction of $\tilde{\mathcal{G}}$ is designed to preserve safety-relevant attributes (DoS, distance, slope, traffic factor, flow, and exposure), so that the clustered network remains comparable in safety terms to the original. We first build a safety-aware weighted adjacency matrix

$$A_{ij} = \begin{cases} w_{ij} \cdot \exp\Big(-\dfrac{d_{ij}}{\gamma}\Big) \cdot \left(\dfrac{\phi_{ij}}{\max\phi}\right)^{\alpha} & \text{if } (i,j)\in\mathcal{A}, \\ 0 & \text{otherwise,} \end{cases} \tag{27}$$

where w_{ij} is DoS, d_{ij} is the spatial distance, ϕ_{ij} is the cyclist flow on arc (i,j), $\gamma > 0$ controls spatial locality and $\alpha \geq 0$ regulates the influence of high-flow arcs. The normalized Laplacian

$$\mathcal{L} = I - D^{-1/2}AD^{-1/2}, \qquad D_{ii} = \sum_j A_{ij},$$

provides a spectral embedding on which we run k-means. Now we proceed to build a constrained k-means, let Φ^{orig} and $\mathcal{E}^{\text{orig}}$ denote the total flow and exposure on $\mathcal{G}$ (cf. (4) and (5)), and let Φ^{clust} and $\mathcal{E}^{\text{clust}}$ be their clustered counterparts defined in (25) and the previous subsection. The clustering is performed by a constrained k-means procedure (Algorithm 1 with pseudo-code in Appendix A) with the following parameters:

i) $\epsilon_{\max}$: maximum allowed DoS variation within any cluster C_r (e.g., $\epsilon_{\max} = 2$ allows clusters of at most two levels of DoS separation),

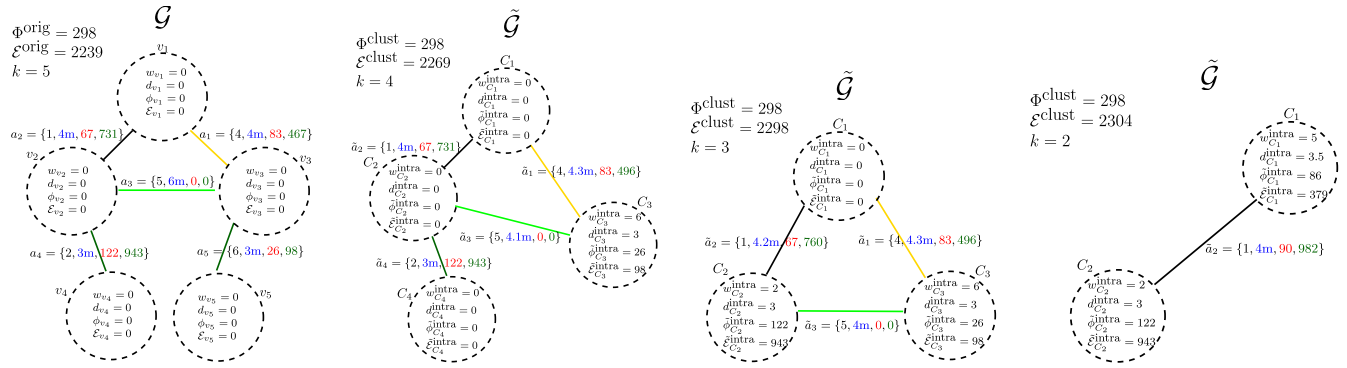
ii) $\delta_{\max}$: maximum spatial diameter of a cluster, making sure that all arcs in the group are relatively close to each other,

iii) $\tau^{(0)}_{\mathcal{E}} := \dfrac{\left|\mathcal{E}^{\text{clust}} - \mathcal{E}^{\text{orig}}\right|}{\mathcal{E}^{\text{orig}}}$: initial tolerance on the relative exposure difference,

iv) $\tau^{(0)}_{\Phi} := \dfrac{\left|\Phi^{\text{clust}} - \Phi^{\text{orig}}\right|}{\Phi^{\text{orig}}}$: initial tolerance on the relative flow difference,

v) α_{relax}: relaxation factor for gradually enlarging $\tau_{\mathcal{E}}$ and τ_{Φ} if needed.

At each iteration, k-means is run in the spectral space with a modified objective that adds Lagrangian penalties λ_1, λ_2 for violating the exposure and flow constraints, while hard constraints enforce DoS homogeneity and bounded spatial diameter inside each cluster. If no feasible solution is found under the current tolerances, the algorithm increases $\tau_{\mathcal{E}}$ and



(a) Original **(b)** Cluster $k = 4$ **(c)** Cluster $k = 3$ **(d)** Cluster $k = 2$

Fig. 2. Algorithm 1 for toy example where $a = \{w_a, d_a(\text{m}), \phi_a, \mathcal{E}_a\}$, $\tilde{a}^{\text{inter}} = \left\{w_{\tilde{a}}^{\text{inter}}, d_{\tilde{a}}^{\text{inter}}(\text{m}), \phi_{\tilde{a}}^{\text{inter}}, \mathcal{E}_{\tilde{a}}^{\text{inter}}\right\}$, and $C^{\text{intra}} = \left\{d_C^{\text{intra}}, d_C^{\text{intra}}(\text{m}), \phi_C^{\text{inter}}, \mathcal{E}_C^{\text{intra}}\right\}$

τ_Φ by a factor $\alpha_{\text{relax}} > 1$ and retries. As a safety valve, if the constrained problem remains infeasible, the method falls back to standard unconstrained k-means in the same spectral embedding, ensuring that a clustering $\mathcal{C} = \{C_1, \ldots, C_k\}$ is always produced. In this way, the clustered network $\tilde{\mathcal{G}}$ approximates the original exposure and flow while strictly controlling intra-cluster safety heterogeneity and spatial extent, which is crucial for fair, safety-aware comparison across cities.

A. Mathematical Guarantees

We start by demonstrating that the Laplacian defined, is indeed a Laplacian:

Theorem 1. *The matrix $\mathcal{L} = I - D^{-1/2}AD^{-1/2}$ is a valid symmetric normalized graph Laplacian.*

Proof. The proof is shown in Appendix A. □

Parameter Determination: The parameters $\gamma > 0$ and $\alpha \geq 0$ in Eq. (27) control the influence of spatial distance and flow magnitude, respectively, on the adjacency weights. They must be chosen to ensure the resulting graph structure reflects meaningful connectivity for clustering. The proof of the Laplacian's properties holds for any $\gamma > 0$ and $\alpha \geq 0$. Therefore, the Laplacian $\mathcal{L}$ maintains its fundamental properties for any admissible choice of γ and α. These parameters should be calibrated based on the specific network and desired clustering outcome, for instance, by ensuring the eigenvalues of $\mathcal{L}$ exhibit a clear gap to suggest a natural number of clusters k.

Theorem 2 (Flow Preservation in Network Clustering). *Given a di-graph $\mathcal{G} = (\mathcal{V}, \mathcal{A}, \mathcal{W})$ with demand matrix $q_{(u,v)}$ and its clustered representation $\tilde{\mathcal{G}} = (\tilde{\mathcal{V}}, \tilde{\mathcal{A}}, \tilde{\mathcal{W}})$ constructed via Algorithm 1, the total network flow is preserved:*

$$\Phi^{orig} \triangleq \sum_{a \in \mathcal{A}} \phi_a = \sum_{\tilde{a} \in \tilde{\mathcal{A}}} \tilde{\phi}_{\tilde{a}} = \Phi^{clust}.$$

Proof. The proof is shown in Appendix A. □

IV. Network Comparison

The purpose of this work is to enable a fair safety comparison between a cycling network $\mathcal{G}$ and a smaller one $\mathcal{H} = (\mathcal{V}(\mathcal{H}), \mathcal{A}(\mathcal{H}), \mathcal{W}(\mathcal{H}))$ by clustering $\mathcal{G}$ into $\tilde{\mathcal{G}} = (\mathcal{C}, \tilde{\mathcal{A}}, \tilde{\mathcal{W}})$ such that $|\mathcal{V}(\mathcal{H})| = |\mathcal{C}| =: n$. For each node $v_i \in \mathcal{V}(\mathcal{H})$ and cluster $C_r \in \mathcal{C}$ we define a node-exposure weight by aggregating incident exposure. This size matching enables meaningful comparison between cities, super index $\tilde{\mathcal{G}}$ and $\mathcal{H}$ will be write to indicate to which network we are referring to.

We introduce two complementary metrics: i) a Gromov–Wasserstein distance that compares the intrinsic structure of cyclist exposure, and iii) an exposure-weighted adjacency kernel distance inspired by graphon comparison.

i) *Gromov–Wasserstein (GW) metric*: For each directed arc $(i, j) \in \mathcal{A}(\mathcal{H})$, we define the cyclist exposure

$$\mathcal{E}_{ij} = \phi_{ij} \cdot f_{ij} \cdot d_{ij}, \tag{28}$$

where ϕ_{ij} is the cyclist flow on arc (i, j), f_{ij} is the traffic factor (which depends on the local degree of separation, AADT, etc.), and d_{ij} is the arc length. We assign an exposure "mass" to each node $v_i \in \mathcal{V}(\mathcal{H})$ by summing the exposure of all incident arcs:

$$\mu^{\mathcal{H}}(v_i) = \sum_{(i,j) \in \mathcal{A}(\mathcal{H})} \mathcal{E}_{ij} + \sum_{(j,i) \in \mathcal{A}(\mathcal{H})} \mathcal{E}_{ji}. \tag{29}$$

For the clustered network $\tilde{\mathcal{G}}$, each superarc $(C_r, C_s) \in \tilde{\mathcal{A}}^{\mathcal{G}}$ has an analogous exposure

$$\tilde{\mathcal{E}}_{rr'} = \tilde{\phi}_{rr'}^{\text{inter}} \cdot f_{rr'}^{\text{inter}} \cdot d_{rr'}^{\text{inter}}, \tag{30}$$

where $\tilde{\phi}_{rr'}^{\text{inter}}$ is the aggregated inter-cluster flow, $f_{rr'}^{\text{inter}}$ is the inter-cluster traffic factor derived from the minimum degree of separation between the two clusters, and $d_{rr'}^{\text{inter}}$ is the flow-weighted inter-cluster distance.

We then define the node mass of each cluster node (supernode) $C_r \in \tilde{\mathcal{V}}^{\mathcal{G}}$ as

$$\mu^{\tilde{\mathcal{G}}}(C_r) = \sum_{(C_r, C_s) \in \tilde{\mathcal{A}}^{\mathcal{G}}} \tilde{\mathcal{E}}_{rr'} + \sum_{(C_s, C_r) \in \tilde{\mathcal{A}}^{\mathcal{G}}} \tilde{\mathcal{E}}_{r'r} + \tilde{\mathcal{E}}_{C_r}^{\text{intra}}, \tag{31}$$

where $\tilde{\mathcal{E}}^{\text{intra}}_{C_r}$ denotes the intra-cluster exposure accumulated within C_r. We normalize these node masses into probability measures:

$$\hat{\mu}^{\mathcal{H}}(v_i) = \frac{\mu^{\mathcal{H}}(v_i)}{\sum_{u\in\mathcal{V}(\mathcal{H})}\mu^{\mathcal{H}}(u)}, \qquad \hat{\mu}^{\tilde{\mathcal{G}}}(C_r) = \frac{\mu^{\tilde{\mathcal{G}}}(C_r)}{\sum_{s\in\tilde{\mathcal{V}}^{\mathcal{G}}}\mu^{\tilde{\mathcal{G}}}(C_s)}. \tag{32}$$

These distributions describe how cyclist exposure is spatially allocated across each network. To compare the exposure structure without requiring an explicit node-to-node correspondence, we use the GW distance. This requires pairwise dissimilarity matrices for each network.

Let $x_i \in \mathbb{R}^2$ denote the geographical coordinates (e.g., projected planar coordinates) of node $v_i \in \mathcal{V}(\mathcal{H})$, and let $c_r \in \mathbb{R}^2$ denote the centroid coordinates of the supernode $C_r \in \tilde{\mathcal{V}}^{\mathcal{G}}$. We define the intra-network dissimilarities

$$D^{\mathcal{H}}_{ij} = \left|\hat{\mu}^{\mathcal{H}}(v_i) - \hat{\mu}^{\mathcal{H}}(v_j)\right| + \gamma\,\|x_i - x_j\|_2, \quad \forall i,j \in \mathcal{V}(\mathcal{H}), \tag{33}$$

and

$$D^{\tilde{\mathcal{G}}}_{rr'} = \left|\hat{\mu}^{\tilde{\mathcal{G}}}(C_r) - \hat{\mu}^{\tilde{\mathcal{G}}}(C_s)\right| + \gamma\,\|c_r - c_s\|_2, \quad \forall r,s \in \tilde{\mathcal{V}}^{\mathcal{G}}, \tag{34}$$

where $\gamma > 0$ balances exposure dissimilarity and geographic separation. And now, let $\Pi(\hat{\mu}^{\mathcal{H}}, \hat{\mu}^{\tilde{\mathcal{G}}})$ be the set of admissible transport couplings $\pi \in \mathbb{R}_+^{k\times k}$ such that

$$\sum_{s=1}^{k}\pi_{is} = \hat{\mu}^{\mathcal{H}}(v_i), \qquad \sum_{i=1}^{k}\pi_{is} = \hat{\mu}^{\tilde{\mathcal{G}}}(C_s). \tag{35}$$

The squared GW distance is then defined as

$$GW^2(\mathcal{H},\tilde{\mathcal{G}}) = \min_{\pi\in\Pi(\hat{\mu}^{\mathcal{H}},\hat{\mu}^{\tilde{\mathcal{G}}})} \sum_{i,j\in\mathcal{V}(\mathcal{H})}\sum_{r,r'\in\tilde{\mathcal{V}}^{\mathcal{G}}} \left(D^{\mathcal{H}}_{ij} - D^{\tilde{\mathcal{G}}}_{rr'}\right)^2 \pi_{ir}\,\pi_{jr'}, \tag{36}$$

and the GW distance is

$$GW(\mathcal{H},\tilde{\mathcal{G}}) = \sqrt{GW^2(\mathcal{H},\tilde{\mathcal{G}})}. \tag{37}$$

A small value of $GW(\mathcal{H},\tilde{\mathcal{G}})$ indicates that high-exposure, structurally important regions in the smaller network can be consistently matched to analogous regions in the clustered larger network, in terms of both spatial embedding and relative exposure load.

V. Benchmark Example

We illustrate the proposed comparison framework on the toy example in Fig. 3. The original small network is denoted by $\mathcal{H}$, and the larger network $\mathcal{G}$ is compressed into a clustered representation $\tilde{\mathcal{G}}$ with $k = 3$ clusters, following the same clustering configuration as in Fig. 2. In this example, $\tilde{\mathcal{G}}$ has roughly half as many nodes as $\mathcal{G}$ while preserving the key safety attributes (flow, exposure, DoS) in aggregated form at the cluster level.

We first compute the Wasserstein distance between the node–exposure measures of $\mathcal{H}$ and $\tilde{\mathcal{G}}$, obtaining $W = 0.412$. In our normalized setting, this value indicates that the spatial allocation of cyclist exposure in the clustered network remains close to that of the original network, despite the substantial reduction in network size. Second, we evaluate

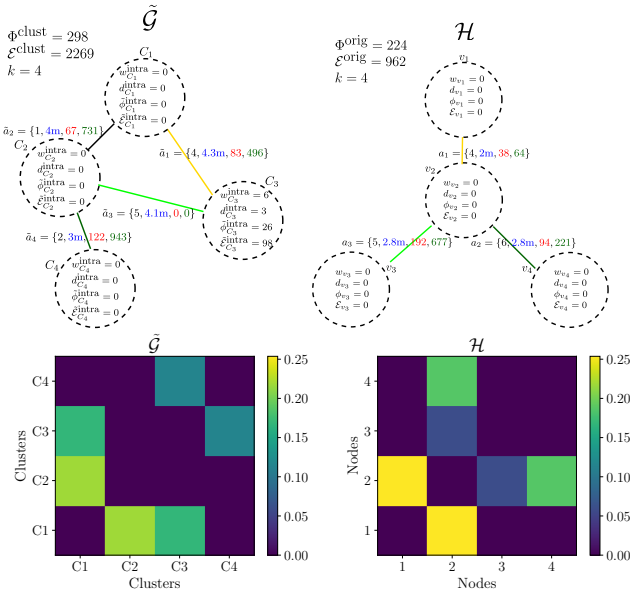


Fig. 3. Kernel graph process with $k = 4$

the exposure–weighted adjacency kernel distance $d_{\text{ker}}(\mathcal{H},\tilde{\mathcal{G}})$, which summarises the mean absolute difference between the normalized exposure kernels of the two networks. For this example we obtain $d_{\text{ker}} = 7.07 \times 10^{-2}$. The corresponding normalized kernels are visualized in the bottom row of Fig. 3, where the colour scale reflects the exposure-weighted strength of connections between node (or cluster) pairs. Both heat maps exhibit a similar block structure and comparable high-intensity entries on and near the diagonal, showing that exposure-weighted connectivity between highly exposed regions is largely preserved. Taken together, the Wasserstein distance captures how total exposure mass is redistributed across space, while the kernel-based distance assesses how risky corridors connect regions within each network. In this benchmark, they jointly indicate that the clustered representation $\tilde{\mathcal{G}}$ is a faithful and size-reduced surrogate of $\mathcal{H}$ for safety comparison.

VI. Case Study

We illustrate the proposed framework on three French cities with well-developed cycling networks: Lyon (11 857 nodes), Strasbourg (11 417 nodes), and Grenoble (6 773 nodes). Grenoble is the smallest network and is taken as the reference. For each of the larger cities, we apply Algorithm 1 to cluster their network down to $|\mathcal{C}| = 6\,773$ supernodes, yielding clustered networks $\tilde{\mathcal{G}}^{\text{Lyon}}$ and $\tilde{\mathcal{G}}^{\text{Strasbourg}}$ that are comparable in size to $\mathcal{H}^{\text{Grenoble}}$. We then compare each clustered network to Grenoble using the Wasserstein and kernel–graphon metrics introduced in Section IV.

Figure 4 shows how the total clustered exposure $\mathcal{E}^{\text{clust}}$ (bars, left axis) and total clustered flow Φ^{clust} (red dashed line, right axis) evolve as the number of clusters decreases, for Lyon (top) and Strasbourg (bottom). The blue vertical line marks the point where the number of clusters equals the number of Grenoble nodes (6 773), while the red vertical line indicates the onset of noticeable divergence between the original and

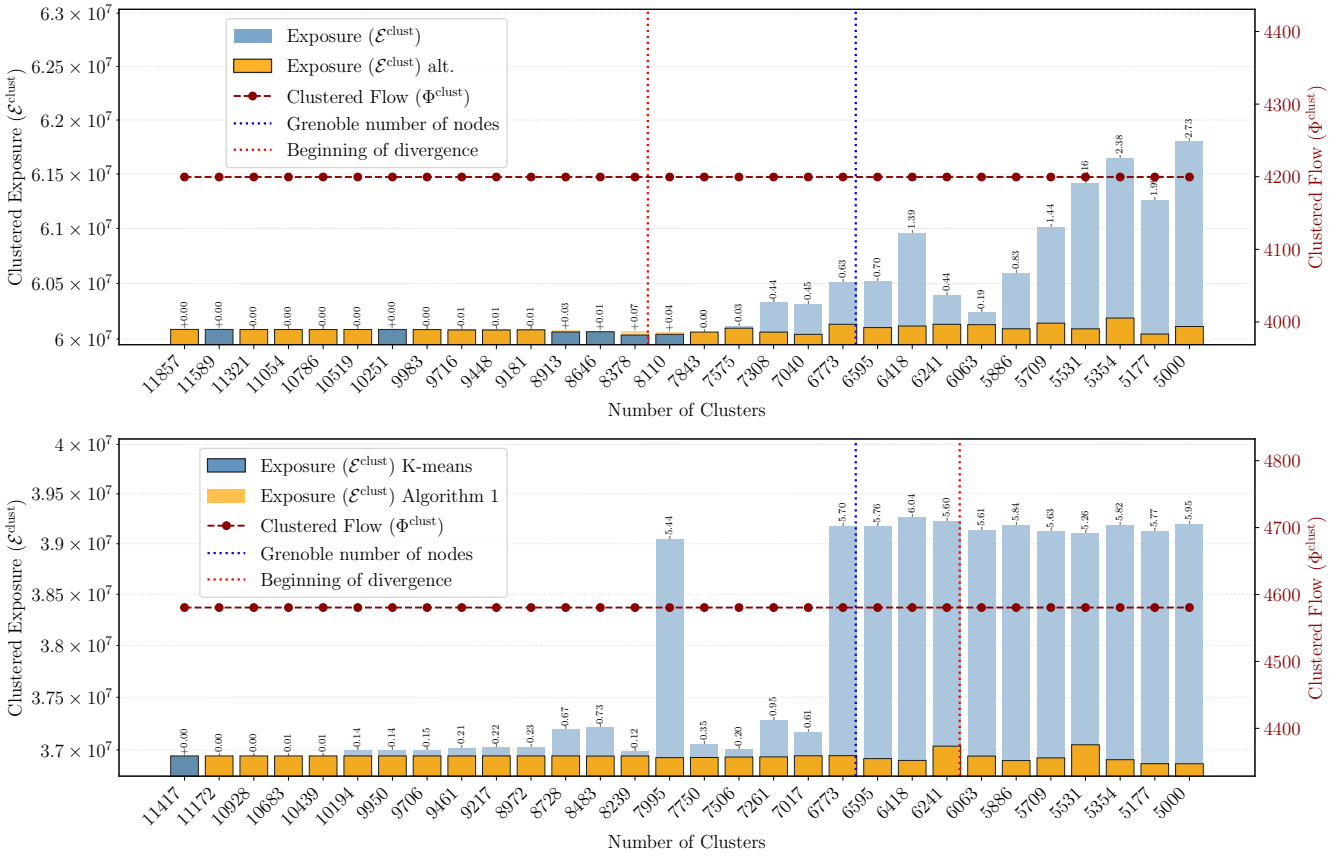


Fig. 4. Evolution of cyclist exposure and cyclist flow through the Algorithm A, on the top Lyon and on the bottom Strasbourg case.

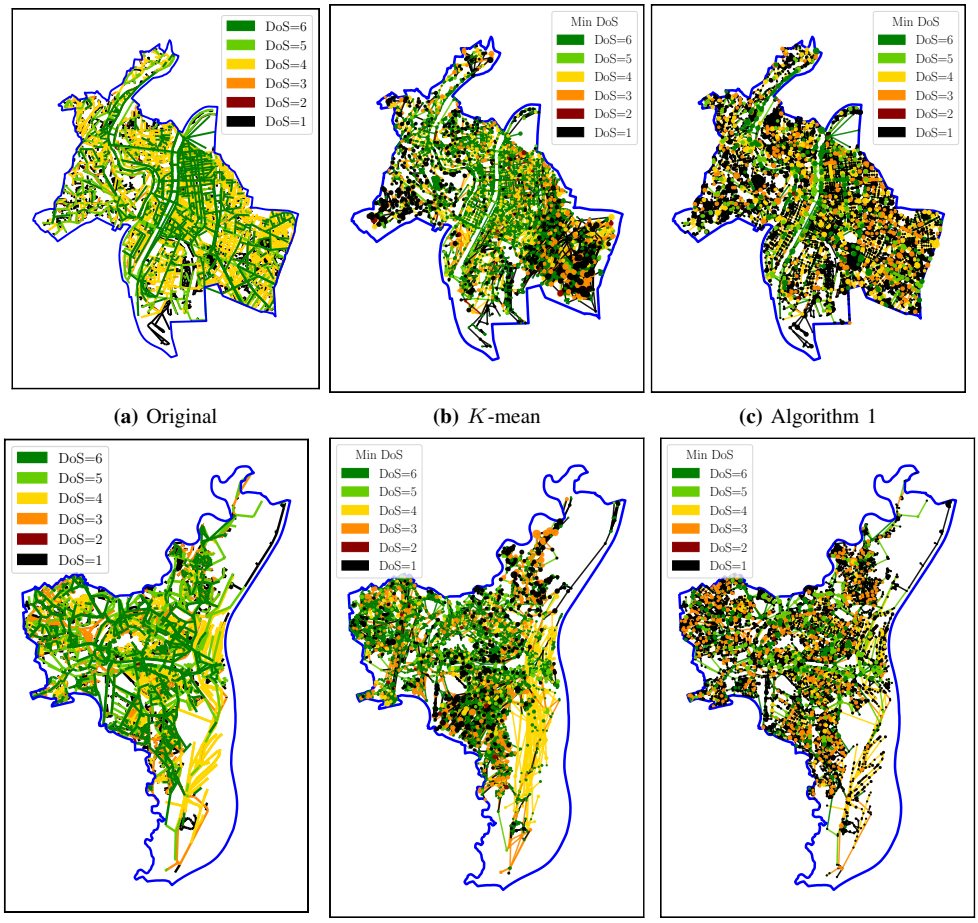
clustered exposure. Two clustering strategies are compared: unconstrained k-means and Algorithm 1. Across both cities, Algorithm 1 consistently produces clustered networks whose total exposure remains much closer to the original value, particularly in the high-compression regime (towards the right of the plots). Lyon reaches the divergence threshold earlier: once the number of clusters falls below about 6 200, the exposure under k-means and (to a lesser extent) Algorithm 1 deviates more strongly from the original network. In contrast, Strasbourg maintains a near-constant exposure over a wider range of cluster counts, suggesting that its DoS structure can be compressed more aggressively without losing safety-relevant information. The clustered flow remains remarkably stable under Algorithm 1, confirming that the total cyclist demand is well preserved during clustering.

Figure 5 compares the spatial distribution of the degree of separation before and after clustering, for Lyon (top row) and Strasbourg (bottom row). The first column shows the original networks coloured by DoS, the second the result of unconstrained k-means, and the third the result of Algorithm 1. While both methods reduce the number of nodes to match Grenoble, the unconstrained k-means clustering tends to mix arcs of very different DoS within the same cluster, leading to patches where risky and safe segments are aggregated and the minimum DoS is not systematically preserved. Algorithm 1, by construction, assigns to each cluster the worst (minimum) DoS of its constituent arcs and enforces bounds on intra-cluster DoS and spatial diameter. Visually, this yields clustered networks whose high-risk corridors and low-risk greenways remain clearly aligned with those of the original networks, even under substantial node reduction. This qualitative behaviour is consistent with the exposure and flow curves in Figure 4.

To inspect the exposure structure more locally, Figure 6 shows heatmaps of the exposure kernels restricted to the 100 most exposed nodes in each network (after log-rescaling for visibility). The top row compares Grenoble to clustered Lyon, and the bottom row Grenoble to clustered Strasbourg. In each panel, bright pixels off the diagonal correspond to pairs of high-exposure nodes that are strongly connected by risky corridors. Both comparisons exhibit a few dense blocks along and near the diagonal, reflecting clusters of nearby high-exposure nodes. However, the pattern for Grenoble–Strasbourg is visually more similar: the main blocks and their relative positions in the matrix align more closely than in the Grenoble–Lyon case, where the high-exposure structure appears more dispersed. This visual impression is reflected in the kernel–graphon distance:

$$D_{\text{ker}}(\mathcal{H}^{\text{Grenoble}}, \tilde{\mathcal{G}}^{\text{Strasbourg}}) = 4.36 \times 10^{-8},$$
$$D_{\text{ker}}(\mathcal{H}^{\text{Grenoble}}, \tilde{\mathcal{G}}^{\text{Lyon}}) = 4.365 \times 10^{-8},$$

with Strasbourg achieving a slightly smaller discrepancy at numerical precision. Figure 7 displays the exposure-based node strength for Grenoble, clustered Lyon, and clustered Strasbourg, where each node is ordered by geographic po-



(a) Original **(b)** K-mean **(c)** Algorithm 1

Fig. 5. Clustering process with Algorithm 1, on the first column we have the original networks, then the cluster one with K-mean and finally with Algorithm 1.

sition (longitude–latitude) and the strength is normalized to $[0, 1]$ using a common nonlinear rescaling. All three networks show a characteristic pattern of many low-exposure nodes punctuated by sharp peaks corresponding to major high-exposure junctions or corridors. The overall silhouettes of Grenoble and clustered Strasbourg are more alike: high peaks tend to appear in similar longitudinal bands, indicating that heavy-exposure zones occupy comparable parts of the urban fabric. Clustered Lyon, while still showing similar qualitative behaviour, exhibits peaks that are more concentrated in a few broader regions, suggesting a different spatial organisation of its main high-exposure corridors. These differences in node-strength patterns are compatible with both the kernel heatmaps and the Wasserstein distances.

Quantitatively, the Wasserstein distances between node-exposure distributions are

$$GW(\mathcal{H}^{\text{Grenoble}}, \tilde{\mathcal{G}}^{\text{Strasbourg}}) = 0.0085,$$
$$GW(\mathcal{H}^{\text{Grenoble}}, \tilde{\mathcal{G}}^{\text{Lyon}}) = 0.0243.$$

The smaller value for Strasbourg indicates that, after clustering to the same number of nodes, its spatial distribution of cyclist exposure is closer to Grenoble's than Lyon's is. Combined with the very small kernel–graphon distances, this suggests that Strasbourg and Grenoble share not only similar global exposure distributions but also similarly structured exposure-weighted connectivity patterns, whereas Lyon deviates more in both respects.

Interestingly, this network-based similarity is coherent with independent survey data from the Baromètre Vélo project [38], where Grenoble and Strasbourg systematically rank among the most bicycle-friendly French cities in terms of perceived safety and comfort, while Lyon, although performing well, tends to score slightly lower. Our framework recovers this qualitative ordering using only infrastructure, flow, and exposure information, showing that clustering-based compression combined with safety-aware metrics can provide a fair and interpretable basis for cross-city comparisons of cycling safety.

VII. Conclusion and Future Work

This paper introduced a safety-aware framework for comparing cycling networks of different sizes by clustering the larger network so that the number of (super)nodes matches that of the smaller one. The clustering algorithm is explicitly designed to preserve key safety attributes—degree of separation, cyclist flow, distance, slope, and angular change—while keeping total exposure and total flow close to their original values. On top of this clustered representation, we proposed several complementary comparison metrics, including a

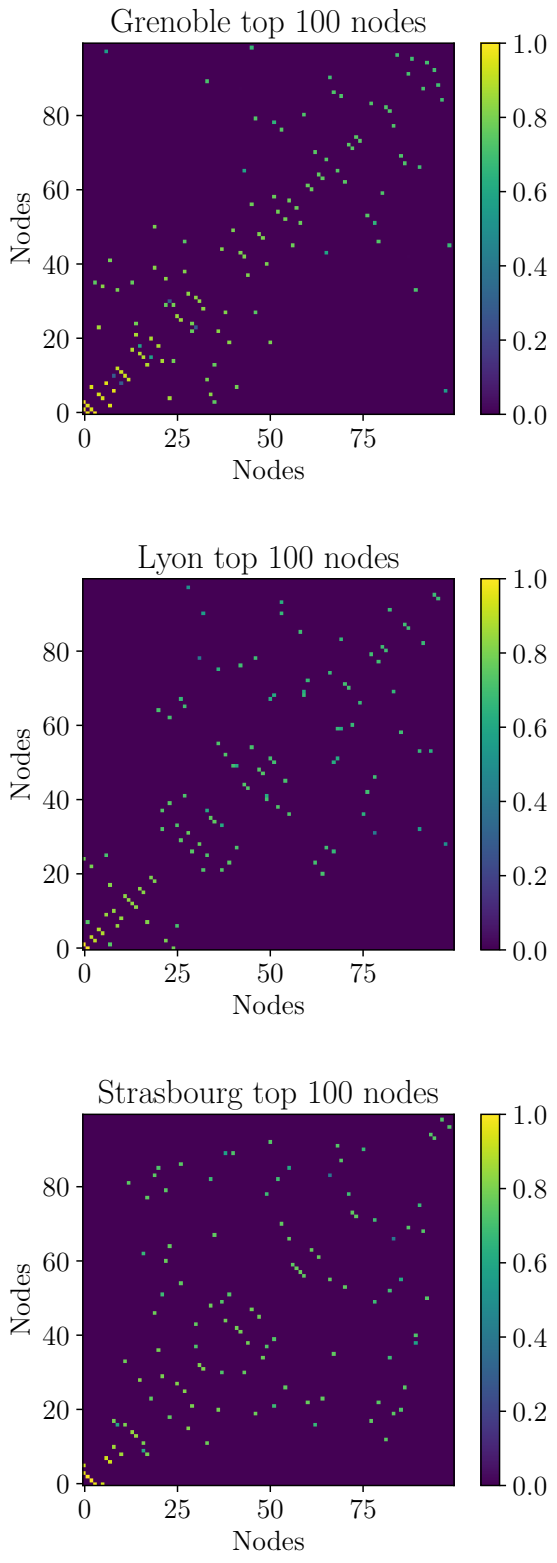


Fig. 6. Heatmaps of top 100 nodes kernels comparison between Grenoble, Lyon and Strasbourg.

Wasserstein distance between node-exposure distributions and an exposure-weighted kernel (graphon-like) distance between adjacency structures. Through a case study on three French cities (Grenoble, Lyon, and Strasbourg), we showed that the proposed clustering method preserves cyclist exposure much better than unconstrained k-means and yields clustered networks that remain topologically close to the originals. The Wasserstein and kernel distances reveal that, once networks are brought to the same size, Strasbourg is structurally more similar to Grenoble than Lyon is, both in terms of the spatial distribution of exposure and in terms of exposure-weighted connectivity. These findings are consistent with independent perception-based indicators (Baromètre Vélo), suggesting that our purely infrastructure- and flow-based metrics capture meaningful aspects of cycling safety and comfort.

The present study focuses on benchmarking and comparison of existing networks. An important next step is to exploit the clustered representation for optimization. In earlier work [39] we proved that the underlying network design problem is NP-hard and currently rely on a genetic algorithm operating directly on the original graph to approach the global optimum. This quickly becomes computationally demanding on large networks.

A natural direction is therefore to formulate the same safety-oriented network design problem on the clustered network $\tilde{\mathcal{G}}$ (e.g., selecting cluster-level upgrades, reclassifications, or new protected links under budget and equity constraints), then, map optimized cluster-level decisions back to the original network, and empirically quantify how close the resulting objective values (exposure, flow, fairness) are to those obtained by optimizing directly on the full graph and use this analysis to design strategies where optimization alternates between coarse (cluster-level) and fine (original-level) representations, in order to better approximate the global optimum at a fraction of the computational cost.

Beyond this, our framework opens several additional research avenues: integrating time-dependent flows and crash data to study temporal robustness; extending the methodology to multi-modal networks (cycling–public transport) and equity-aware objectives; and exploring alternative clustering schemes or OT-based distances tailored to policy questions such as corridor prioritization or phased investment planning. Ultimately, we aim to turn the proposed clustering and comparison tools into a decision-support pipeline that allows cities to both benchmark their cycling networks and explore cost-effective, safety-improving interventions at scale.

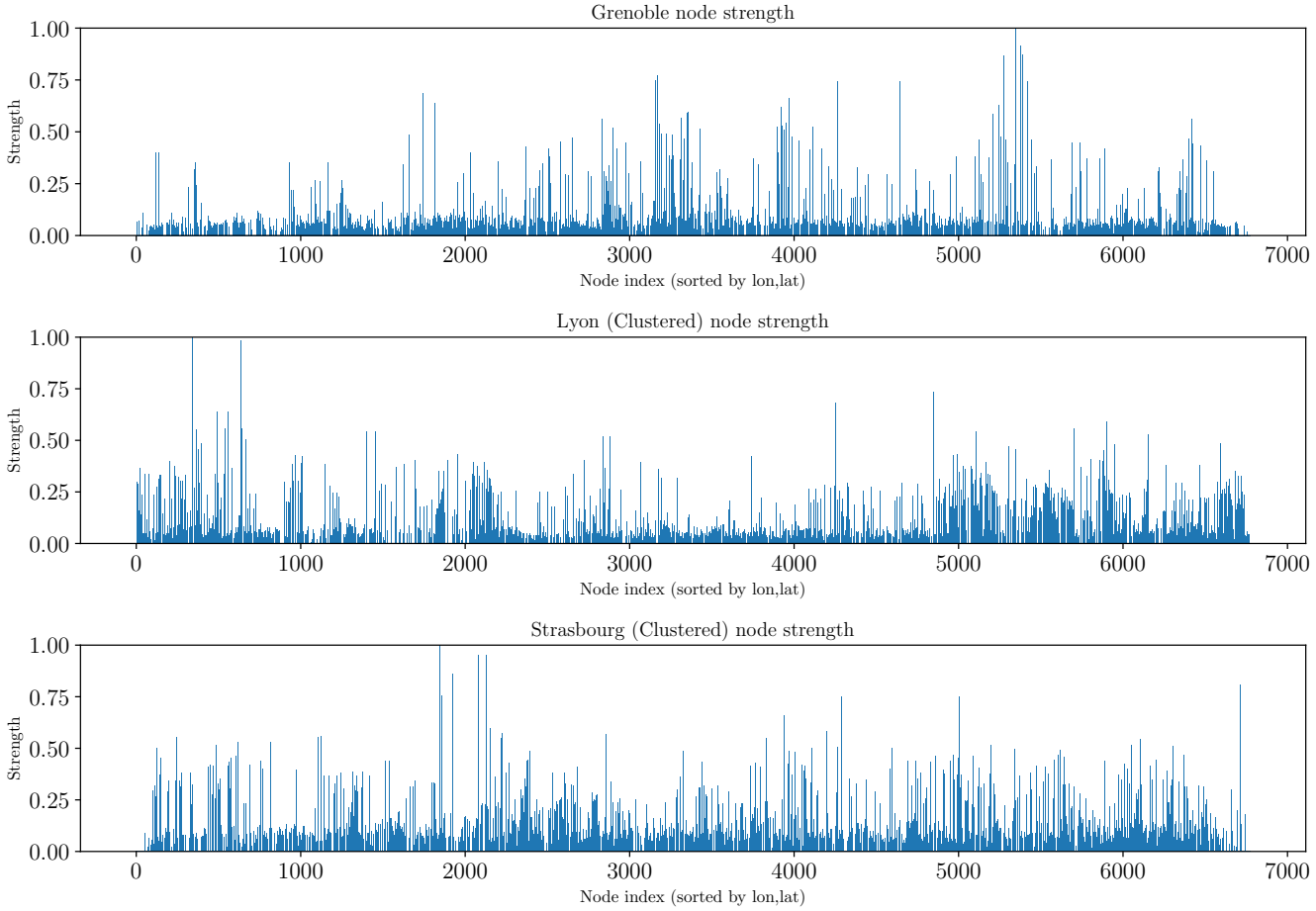


Fig. 7. Node strength of Grenoble and clustered Lyon and Strasbourg cycling networks.

APPENDIX A

A. Algorithm 1 Pseudo-code

Algorithm 1 Constrained Spectral Clustering for Bike Networks

Require: $\mathcal{G} = (\mathcal{V}, \mathcal{A}, \mathcal{W})$, k, $\epsilon_{\max}$, $\delta_{\max}$, $\tau_{\mathcal{E}}^{(0)}$, $\tau_{\Phi}^{(0)}$, α_{relax}, λ_1, λ_2, $T_{\max}$.

Ensure: $\tilde{\mathcal{G}} = (\tilde{\mathcal{V}}, \tilde{\mathcal{A}}, \tilde{\mathcal{W}})$

Spectral Embedding:

$$A_{ij} = \begin{cases} w_{ij} \cdot e^{-d_{ij}/\gamma} \cdot \left(\frac{\phi_{ij}}{\max \phi}\right)^{\alpha}, & (i,j) \in \mathcal{A} \\ 0, & \text{otherwise} \end{cases}$$

$$\mathcal{L} = I - D^{-1/2} A D^{-1/2}, \quad D_{ii} = \sum_j A_{ij}$$

Compute first m eigenvectors of $\mathcal{L}$ to obtain $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times m}$.

Initialize: $\tau_{\mathcal{E}} \leftarrow \tau_{\mathcal{E}}^{(0)}, \quad \tau_{\Phi} \leftarrow \tau_{\Phi}^{(0)}$

for $t = 1, \ldots, T_{\max}$ **do**

For each $v_i \in \mathcal{V}$:

$$r^* = \arg \min_{r \in \{1,\ldots,k\}} \left\{ \|\mathbf{x}_i - \mu_r\|^2 + \lambda_1 \cdot \mathbb{I}_{\text{DoSViol}} + \lambda_2 \cdot \mathbb{I}_{\text{DistViol}} \right\}$$

subject to:

$$\max_{v_j \in C_r} \left| w_{ij} - \min_{v_\ell \in C_r} w_{i\ell} \right| \le \epsilon_{\max}, \quad \max_{v_j \in C_r} d_{ij} \le \delta_{\max}$$

Assign $v_i \to C_{r^*}$ and update $\mu_r \leftarrow \frac{1}{|C_r|} \sum_{v_j \in C_r} \mathbf{x}_j$

Build $\tilde{\mathcal{G}}$ using definitions in Section II-C.

APPENDIX B

Complementary Proofs

Theorem 1 affirms that matrix $\mathcal{L} = I - D^{-1/2} A D^{-1/2}$ is a valid symmetric normalized graph Laplacian.

Proof. The proof follows from the symmetry and non-negativity of the adjacency matrix A and the subsequent definition of $\mathcal{L}$.

i) Symmetry of A: Although $\mathcal{G}$ is directed, the arc attributes are symmetric:

$$w_{(u,v)} = w_{(v,u)}, \quad d_{(u,v)} = d_{(v,u)}, \quad \phi_{(u,v)} = \phi_{(v,u)}.$$

Since A_{ij} is defined as a nonnegative function of $(w_{ij}, d_{ij}, \phi_{ij})$ with no directional bias, it follows that $A_{ij} = A_{ji}$ and hence $A = A^\top$.

ii) Non-negativity of A: By definition, $w_{ij} \ge 0$, $d_{ij} \ge 0$, $\phi_{ij} \ge 0$, $\max(\boldsymbol{\phi}) > 0$, and the exponential function is always positive. Therefore, for all $(i,j) \in \mathcal{A}$, $A_{ij} \ge 0$. For non-adjacent nodes, $A_{ij} = 0$ by definition.

iii) Properties of $\mathcal{L}$: Consider the (unnormalized) Laplacian $L = D - A$. For any $\mathbf{y} \in \mathbb{R}^{|\mathcal{V}|}$,

$$\mathbf{y}^\top L \mathbf{y} = \frac{1}{2} \sum_{i,j} A_{ij} (y_i - y_j)^2 \ \ge \ 0,$$

so L is positive semidefinite. The normalized Laplacian can be written as

$$\mathcal{L} \;=\; D^{-1/2} L D^{-1/2}.$$

For any $\mathbf{x}$, let $\mathbf{y} = D^{-1/2}\mathbf{x}$. Then

$$\mathbf{x}^\top \mathcal{L}\mathbf{x} = (D^{-1/2}\mathbf{x})^\top L (D^{-1/2}\mathbf{x}) = \mathbf{y}^\top L \mathbf{y} \;\geq\; 0,$$

which shows that $\mathcal{L}$ is positive semidefinite. Since $\mathcal{L}$ is also symmetric, all its eigenvalues are real and nonnegative (and, by standard results, lie in $[0, 2]$). This completes the proof. □

Theorem 2 guarantees that given a di-graph $\mathcal{G} = (\mathcal{V}, \mathcal{A}, \mathcal{W})$ with demand matrix $q_{(u,v)}$ and its clustered representation $\tilde{\mathcal{G}} = (\tilde{\mathcal{V}}, \tilde{\mathcal{A}}, \tilde{\mathcal{W}})$ constructed via Algorithm 1, the total network flow is preserved:

$$\Phi^{\text{orig}} \triangleq \sum_{a \in \mathcal{A}} \phi_a = \sum_{\tilde{a} \in \tilde{\mathcal{A}}} \tilde{\phi}_{\tilde{a}} = \Phi^{\text{clust}}.$$

Proof. Observe that the clustered demand $\tilde{q}_{(C_r, C_{r'})}$ in (21) aggregates all original demands between nodes in clusters C_r and $C_{r'}$ as

$$\tilde{q}_{(C_r, C_{r'})} = \sum_{\substack{u \in C_r \\ v \in C_{r'}}} q_{(u,v)},$$

ensuring total demand preservation

$$\sum_{u,v \in \mathcal{V}} q_{(u,v)} = \sum_{C_r, C_{r'} \in \tilde{\mathcal{V}}} \tilde{q}_{(C_r, C_{r'})}.$$

In the original graph $\mathcal{G}$, the all-or-nothing assignment (4) yields arc flows

$$\phi_a = \sum_{u,v} q_{(u,v)} \cdot \mathbb{I}_{a \in p_{(u,v)}},$$

while in the clustered graph $\tilde{\mathcal{G}}$, (22) gives superarc flows

$$\tilde{\phi}_{\tilde{a}} = \sum_{C_r, C_{r'}} \tilde{q}_{(C_r, C_{r'})} \cdot \mathbb{I}_{\tilde{a} \in \tilde{p}_{(C_r, C_{r'})}}.$$

The total flow in $\mathcal{G}$ decomposes as:

$$\Phi^{\text{orig}} = \sum_{a \in \mathcal{A}} \phi_a = \sum_{u,v} q_{(u,v)} \underbrace{\sum_{a \in \mathcal{A}} \mathbb{I}_{a \in p_{(u,v)}}}_{=1} \tag{38}$$

$$= \sum_{u,v} q_{(u,v)}, \tag{39}$$

since each path $p_{(u,v)}$ covers exactly one set of arcs. Similarly, for $\tilde{\mathcal{G}}$:

$$\Phi^{\text{clust}} = \sum_{\tilde{a} \in \tilde{\mathcal{A}}} \tilde{\phi}_{\tilde{a}} = \sum_{C_r, C_{r'}} \tilde{q}_{(C_r, C_{r'})} \underbrace{\sum_{\tilde{a} \in \tilde{\mathcal{A}}} \mathbb{I}_{\tilde{a} \in \tilde{p}_{(C_r, C_{r'})}}}_{=1} \tag{40}$$

$$= \sum_{C_r, C_{r'}} \tilde{q}_{(C_r, C_{r'})}, \tag{41}$$

where the indicator sums equal 1 because each clustered path $\tilde{p}_{(C_r, C_{r'})}$ traverses exactly one sequence of superarcs. By demand conservation, we conclude $\Phi^{\text{orig}} \equiv \Phi^{\text{clust}}$. □

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